THE BIVARIANT ACTION OF A CLOSED SUBSCHEME FAMILY ON A CYCLE IS THE CYCLE OF THE FIBRED PRODUCT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file relative-cycles.tex; no statement or proof step was rewritten, reordered or removed. The counted span is the author environment and its own proof environment. References inside the span are handed to the assembler as navigation stubs that carry the canonical tag of the cited tag only; whether each of them is load-bearing is recorded on the stub itself.

Allowed standard prerequisites (editorial).

- Cycle family associated to a closed subscheme (relative-cycles chapter, Example example-family-associated-closed, canonical tag 0H4S). cited at statement line 2435. Defines the family alpha = [Z/X/Y]_r.
- Cycle associated to a closed subscheme (Chow Homology, Definition chow-definition-cycle-associated-to-closed-subscheme, canonical tag 02QU). cited at statement lines 2437-2438. Defines the prime cycle beta = [W]_e.
- Coherence of the bivariant action (relative-cycles chapter, Lemma lemma-action-coherent, canonical tag 0H69). cited at proof lines 2443-2444. Specialized to F = O_Z and G = O_W: the action of the module family of O_Z on the module family of O_W is the module family of the tensor product.

Source reference. Original reference 0H4S: sources/stacks-chow/tags.txt, source lines 21362–21362. Cited at statement line 2435: defines alpha = [Z/X/Y]_r.

Source reference. Original definition 02QU: sources/stacks-chow/tags.txt, source lines 3601–3601. Cited at statement lines 2437-2438: defines beta = [W]_e.

Source reference. Original lemma 0H69: sources/stacks-chow/tags.txt, source lines 21414–21414. Cited at proof lines 2443-2444: the special case F = O_Z, G = O_W.

Lemma 0.2. *Let (S, δ) and $f : X \to Y$ be as above. Let $Z \subset X$ be a closed subscheme of relative dimension $\leq r$ over Y. Set $\alpha = [Z/X/Y]_r$ (Example 0H4S). Let $W \subset Y$ be a closed subscheme of δ-dimension $\leq e$. Set $\beta = [W]_e$ (Chow Homology, Definition 02QU). If Z is flat over Y, then $\alpha \cap \beta = [Z \times_Y W]_{r+e}$.*

Proof. This is a special case of Lemma 0H69 if we take $\mathcal{F} = \mathcal{O}_Z$ and $\mathcal{F} = \mathcal{O}_W$. □

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